

Meta-analyses

The impact of dietary fiber consumption on human health: An umbrella review of evidence from 17,155,277 individuals



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SUMMARY

Background: Dietary fiber plays a crucial role in disease prevention through multiple protective mechanisms. While previous reviews have examined its health effects, recent advances necessitate an updated synthesis of the evidence.

Objectives: This umbrella review systematically evaluates meta-analyses of observational studies to assess the strength and credibility of associations, and the quality of evidence linking dietary fiber intake to diverse health outcomes, providing a comprehensive assessment of these associations.

Methods: We updated our previous umbrella review by searching major databases (October 21, 2017–December 1, 2024) for meta-analyses on dietary fiber and health outcomes. Methodological quality was assessed using AMSTAR-2 (A Measurement Tool to Assess Systematic Reviews), and

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evidence credibility was classified based on predefined criteria, accounting for heterogeneity (I^2), small-study effects, and excess significance bias.

Results: Thirty-three meta-analyses covering 38 health outcomes and 17,155,277 individuals were included. Of these, 29 (76 %) reported significant inverse associations ($p < 0.05$) between higher fiber intake and disease risk. Convincing evidence (Class I) was identified for cardiovascular disease (CVD) mortality, pancreatic cancer, and diverticular disease. Highly suggestive evidence (Class II) supported associations with all-cause mortality, CVD, coronary heart disease and ovarian cancer. Suggestive evidence (Class III) was observed for 16 outcomes, while six outcomes (16 %) exhibited weak evidence (Class IV). Quality assessment indicated that six meta-analyses were of high methodological quality, while others had some methodological limitations.

Conclusions: Higher dietary fiber intake is associated with lower risk of multiple chronic diseases, particularly CVD mortality, pancreatic cancer, and diverticular disease. Despite these benefits, intake remains below recommended levels worldwide. Public health strategies should promote fiber-rich diets, and future research should further elucidate the role of specific fiber types in disease prevention.

Systematic review registration: <https://osf.io/37tyc/>.

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1. Introduction

Dietary fiber, defined as plant-derived or analogous carbohydrates resistant to digestion or absorption in the small intestine, is a cornerstone of a healthy diet [1,2]. Its physiochemical properties, including bulk volume, viscosity, ion exchange capacity, and fermentation patterns, contribute to metabolic benefits such as improved glucose and lipid regulation and enhanced colonic function [3]. Fiber also alter incretin secretion and modulates gut microbiota, fostering beneficial bacterial shifts that generate potent immune-modulating metabolites [4]. Additionally, pre-clinical studies indicate that fiber reduces markers of inflammation and oxidative stress [5,6], both of which are linked to adverse health outcomes, including cancer and premature mortality [7–11].

Epidemiological evidence consistently supports the health benefits of high dietary fiber intake, with associations to reduced mortality rates, likely by mitigating conditions such as type 2 diabetes, cardiovascular disease (CVD), and gastrointestinal cancers [1]. However, despite the well-documented advantages, fiber intake remains suboptimal in many developed countries [12]. In the UK, for instance, the recommended daily fiber intake is 30 g, yet average consumption remains around 19 g, with no significant improvement since the guidelines were updated in 2015 [13]. Similarly, over 90 % of the Americans fail to meet the 28-g daily recommendation [14]. These trends highlight the ongoing challenge in achieving adequate fiber consumption, underscoring the need for continued public health efforts to promote fiber-rich diets.

Nearly a decade ago, our team published the first umbrella review on dietary fiber impact on human health, finding significant associations between high fiber intake and reduced risks of CVD, pancreatic cancer, and all-cause mortality, though the evidence strength varied by outcome [15]. This updated umbrella review aims to re-evaluate the robustness, consistency, and reliability of recent meta-analyses examining the relationship between dietary fiber intake and health outcomes. While earlier reviews, including our own, have indicated a protective role of dietary fiber against several diseases, they often lacked standardized evidence grading, comprehensive credibility assessments, and inclusion of the most recent studies. By integrating an expanded body of literature, employing rigorous methodological quality assessments (e.g., AMSTAR-2), and stratifying evidence using established credibility criteria, this study directly addresses previous methodological limitations and updates the evidence

base with the latest findings. In doing so, it fills critical knowledge gaps around fiber's role in emerging conditions and helps identify high-priority targets for dietary recommendations and public health interventions.

2. Materials and methods

2.1. Literature search and selection criteria

This umbrella review was conducted following established protocols for systematically evaluating multiple meta-analyses on a specific research topic [16,17]. The protocol was pre-registered at <https://osf.io/37tyc/>.

Our search strategy covered MEDLINE/PubMed, Scopus, and Embase databases, spanning from October 21, 2017 to December 1, 2024. We developed the search strategy based on the PICOS framework (Population, Intervention, Comparison, Outcome, Study design). Briefly, we considered:

- P: any participants or population;
- I: high dietary fiber intake
- C: low dietary fiber intake;
- O: any health outcome;
- S: systematic review followed by meta-analysis.

Keywords and MeSH terms were identified through iterative discussion among the authors and informed by terms used in key articles identified early in the process. The specific search terms used for each database are detailed in [Supplementary Table 1](#). Additionally, we manually reviewed the reference lists of eligible articles for further relevant studies. To assess the comprehensiveness and sensitivity of our search strategy, we conducted a pilot test by verifying that a set of benchmark studies known to be relevant (identified through manual searches and expert knowledge) were successfully retrieved by our search string. We included only formal meta-analyses of observational studies that investigated dietary fiber as a potential factor influencing health outcomes, with no language restrictions. Primary screening, which included title and abstract review, was independently performed by three authors (CDP, CM, LC). The agreement was good across these three researchers ($\kappa = 0.85$). Any discrepancies were resolved through consensus or with a senior author (NV). We implemented several quality control measures, including peer review of the search strategy by two independent reviewers, and use of a standardized data extraction form. Full texts of potentially eligible articles were retrieved, and final eligibility was determined by consensus between two investigators (AC, NV). Eligible

studies were peer-reviewed meta-analyses that focused on longitudinal observational studies (e.g., case-control, retrospective or prospective cohort designs) examining the association between baseline dietary fiber intake and various health outcomes, including CVD, cancer, mortality, obesity/overweight, severe mental illness, diabetes, and metabolic disorders. Health outcomes had to be reported as odds ratios (OR), relative risks (RR), or hazard ratios (HR) at follow-up. We excluded systematic reviews that lacked quantitative synthesis, meta-analyses based on individual data without a systematic review, and meta-analyses that included data from randomized controlled trials.

2.2. Data extraction

Three independent investigators (CDP, CM, LC) performed data extraction for each included systematic review and individual study. The following information was systematically extracted: (1) first author name, (2) publication year, (3) journal, (4) dietary fiber intake and its categorization (e.g., quintiles, quartiles), (5) methods used to ascertain dietary fiber intake (e.g., FFQ, 24-h recall), (6) number of studies included and total sample size, (7) inclusion criteria for study populations, (8) effect size metrics used, (9) study design (case-control or prospective), and (10) numbers of cases (i.e., those having an event) and controls in each study. For each study, we extracted the estimated relative risks (RR, OR, HR), their 95 % confidence intervals (CI), and the number of cases categorized by dietary fiber intake. Assuming a linear association between dietary fiber intake and health outcomes, data for the lowest and highest quantiles were included in the overall analysis. When multiple meta-analyses assessed the same association, we prioritized the meta-analysis with the largest number of studies.

2.3. Assessment of risk of bias

Three authors (LC, CM, CDP) independently rated the methodological quality of the included systematic reviews using the “A MeaSurement Tool to Assess systematic Reviews 2 (AMSTAR 2)” [18]. This tool ranks the quality of a meta-analysis into one of 4 categories, ranging from “critically low” to “high”, based on 16 predefined items [18]. The assessments were then double-checked by another author (NV) to ensure accuracy. For the previous version of the umbrella review, the risk of bias evaluation was conducted de novo, as it was not performed in the earlier iteration of the review.

2.4. Data analysis

We conducted data analysis following the standard methodology for umbrella reviews, as outlined in established frameworks [19,20]. Summary effect sizes and their corresponding 95 % CIs were calculated for each meta-analysis using random-effects models [21]. For meta-analyses that reported continuous measures (e.g., differences in fiber intake across quantiles for body weight), these values were converted into RRs using established conversion formulas in *metaconvert* R package [22]. To account for between-study effects and evaluate the likelihood of consistent results in the new studies, we calculated prediction intervals and their 95 % CIs [23]. Between-study heterogeneity was assessed using the I^2 statistic, with values >50 % indicating high heterogeneity and values >75 % suggesting very high heterogeneity [24]. Small-study effects were evaluated using Egger's regression asymmetry test [25], where a p-value <0.10 was considered indicative of inflated effect sizes in smaller studies [26]. To evaluate excess significance bias, we applied Ioannidis' excess of significance test [27], which compares the observed (O) number of

significant studies with the expected (E) number based on study power. Excess significance was identified when the p-value for the χ^2 -based test was <0.10 [27]. Power estimates for each study were calculated using the largest dataset as the reference for effect size calculations. All the analyses were performed using the R package “metaumbrella” [28].

2.5. Credibility assessment

We classified the evidence from meta-analyses with nominally significant results ($p < 0.05$) into four categories (classes I–IV) based on predefined criteria [29].

- **Convincing (Class I):** Associations with $p < 10^{-6}$, more than 1000 cases (or $>20,000$ participants for continuous outcomes), significant results from the largest component study ($p < 0.05$), 95 % prediction intervals excluding the null, low heterogeneity ($I^2 < 50$ %), and no evidence of small-study effects ($p > 0.10$) or excess significance bias ($p > 0.10$).
- **Highly Suggestive (Class II):** Associations with $p < 0.001$, >1000 cases (or $>20,000$ participants for continuous outcomes), and significant results in the largest component study.
- **Suggestive (Class III):** Associations with $p < 0.01$ and >1000 cases (or $>20,000$ participants for continuous outcomes).
- **Weak (Class IV):** Remaining significant associations with $p < 0.05$.

For ‘Class I’ associations, evidence was re-evaluated after excluding retrospective or case-control studies to address potential reverse causality. All outcomes in this class, except for pancreatic cancer, were based on prospective studies, with pancreatic cancer only having one prospective study included.

3. Results

3.1. Literature review

As shown in Fig. 1, our previous umbrella review identified 18 systematic reviews with meta-analyses eligible for inclusion [30–47]. For this update, we screened 1714 articles across three databases and examined 39 full-text articles (the list of excluded works is reported, with reason, in Supplementary Table 2). From this, 15 new meta-analyses were identified [48–64]. In total, we reviewed 33 systematic reviews with meta-analyses on the relationship between dietary fiber intake and health outcomes in humans.

3.2. Meta-analyses of observational studies

As reported in Table 1, the median number of studies included in meta-analyses was 11 (range, 2–29), with a median of 463,656 participants (total: 17,155,277). The median number of cases across all studies was 11,388. All the meta-analyses, except one (which reports dietary fiber intake as an increase of 7 g/day), compared the highest versus the lowest quantile of dietary fiber intake. In all the meta-analyses included, FFQ was used to ascertain the dietary fiber intake.

The meta-analyses addressed 38 different health outcomes across a broad spectrum (Table 1), including cancer and pre-cancer lesions ($n = 19$), cardiovascular diseases ($n = 5$), gastrointestinal conditions ($n = 4$), metabolic diseases ($n = 3$), and all-cause and specific-cause mortality ($n = 7$). Of the 38 non-overlapping meta-analyses, 29 (76 %) reported nominally significant summary results ($p < 0.05$) (Table 1).

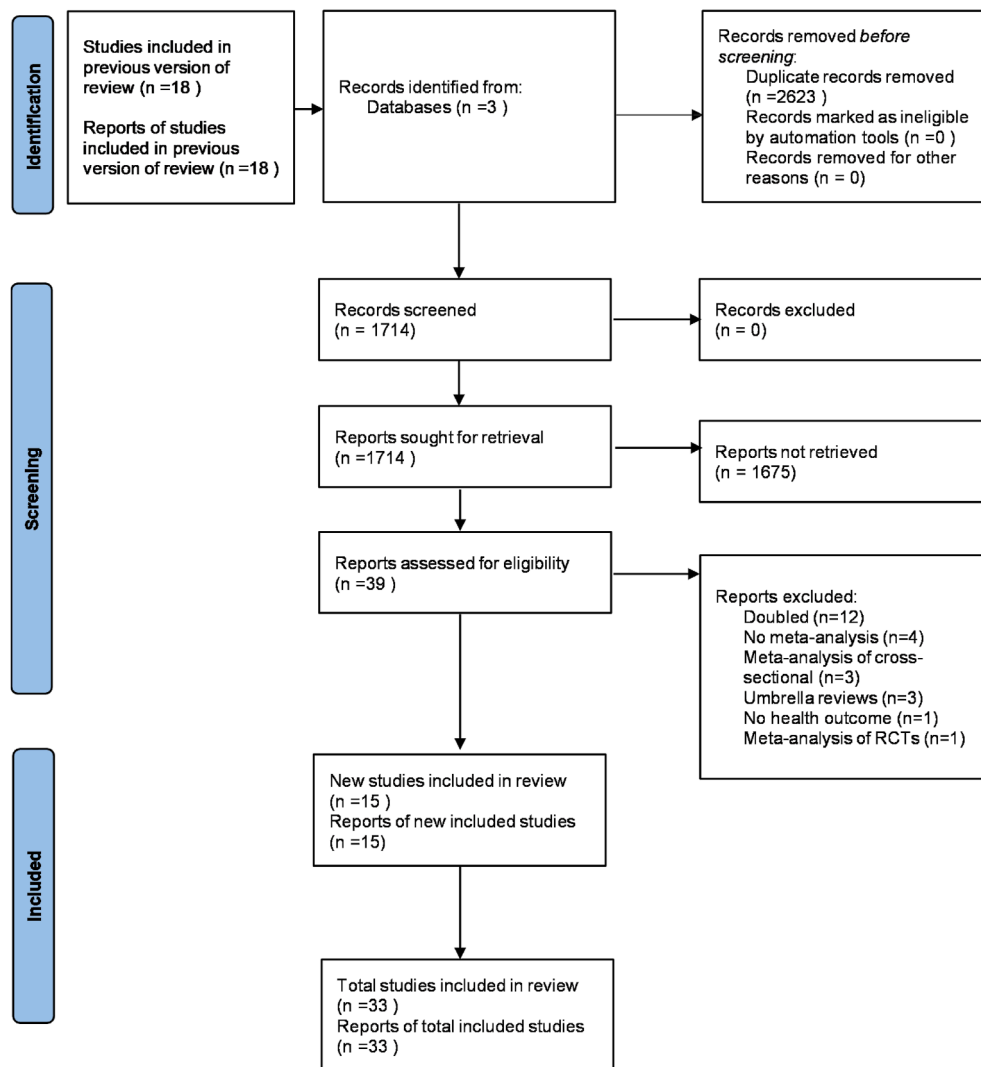


Fig. 1. PRISMA flow-chart.

We applied our evidence classification criteria to assess the strength of the associations. Nine (11 %) meta-analyses demonstrated $P < 10^{-6}$, while another nine (11 %) had a 95 % prediction interval that excluded the null. The majority of studies (79 %, $n = 30$) involved more than 1000 cases (or more than 20,000 participants for continuous fiber intake), 21 meta-analyses (55 %) reported low heterogeneity ($I^2 < 50$ %), and 23 (60 %) had no evidence of small-study effects or excess significant bias. However, publication bias was present in 8 of the 38 meta-analyses (21 %).

Based on these metrics, three (8 %) of the 38 outcomes—all derived from general population studies—demonstrated convincing evidence (Class I), including CVD mortality and the incidence of pancreatic cancer and diverticular disease. Four (11 %) outcomes presented highly suggestive evidence (class II), including all-cause mortality and the incidence of CVD, coronary heart disease, and ovarian cancer. Sixteen (42 %) outcomes, primarily related to cancer, provided suggestive evidence (Class III). Finally, six meta-analyses (16 %) with statistically significant findings were categorized as weak evidence (Class IV), while nine meta-analyses did not report any significant associations between dietary fiber intake and the health outcome of interest (Table 1).

3.3. Quality assessment

The risk of bias was evaluated using the AMSTAR-2 tool, with details provided in [Supplementary Table 3](#). Overall, six works were rated as high quality, while 15 were classified as moderate. Four works were deemed low quality, and the remaining 8 were considered critically low quality.

4. Discussion

This umbrella review provides an updated and comprehensive synthesis of the evidence regarding the association between dietary fiber intake and a range of health outcomes, as summarized in the Infographics (Fig. 2). Our findings reinforce the protective effects of dietary fiber against several chronic diseases. Notably, we identified the strongest evidence for pancreatic cancer, diverticular disease, and CVD mortality, all of which were classified as “convincing” (Class I) based on stringent criteria, including a p -value $< 10^{-6}$, more than 1000 cases, low heterogeneity, and absence of small-study effects or excess significance bias.

This updated umbrella review builds upon our previous synthesis and contributes to the expanding field of nutritional epidemiology by incorporating recent meta-analyses and

Table 1
Health outcomes and evidence class reported in included meta-analyses.

Outcome	Population	Study design included in MA	Level of comparison	N of studies	N of participants	N of cases	Type of metric	Effect size	p-value	I ²	Publication bias	Small study effect or excess significance bias	Excess significance test	95 % CI prediction intervals	Evidence class ^a
Pancreatic cancer [45]	General	Case-control and prospective	Highest vs. lowest category	14	38,141	3287	OR	0.582 (0.501–0.675)	1.18 e ⁻¹²	6.2	No	Neither	No	0.46 0.73	I
CVD mortality [59]	General	Prospective	Highest vs. lowest category	10	806,561	52582	RR	0.818 (0.778–0.861)	8.07e ⁻¹⁵	0	No	Neither	No	0.77 0.87	I
Diverticular disease [48]	General	Prospective	Highest vs. lowest category	5	477,166	19282	RR	0.743 (0.716–0.771)	1.18e ⁻⁵⁵	0	No	Neither	No	0.70 0.789	I
CVD [43]	Mixed	Prospective	Continuous (increase in 7 g/day)	22	2,165,830	25461	RR	0.913 (0.893–0.932)	6.26e ⁻¹⁷	36.7	Yes	Both	No	0.86 0.97	II
All-cause mortality [59]	General	Prospective	Highest vs. lowest category	29	2,363,370	94457	HR	0.762 (0.711–0.816)	9.44e ⁻¹⁵	66.275	No	Neither	No	0.567 1.023	II
Ovarian cancer [53]	Women	Case-control and prospective	Highest vs. lowest category	14	28241	7043	RR	0.771 (0.716–0.83)	5.77e ⁻¹²	0	No	Excess significance bias	Yes	0.710 0.837	II
Coronary heart disease [33]	Not reported	Prospective	Highest vs. lowest category	25	461,187	8591	RR	0.932 (0.906–0.958)	8.10e ⁻⁰⁷	0	Yes	Both	Yes	0.9 0.96	II
Gastric cancer [36]	Mixed	Case-control and prospective	Highest vs. lowest category	26	580,658	6950	OR	0.575 (0.494–0.67)	1.29e ⁻¹²	62.2	Yes	Both	0.34	0.31 1.08	III
Esophageal adenocarcinoma [46]	Mixed	Case-control	Highest vs. lowest category	9	9873	1576	OR	0.495 (0.366–0.67)	5.16e ⁻⁰⁶	67.5	Yes	Excess significance bias	1.00	0.19 1.26	III
Colon cancer [24]	Not reported	Case-control	Highest vs. lowest category	15	9976	3280	OR	0.643 (0.524–0.788)	0.00002	50.4	No	Neither	0.14	0.35 1.16	III
Colorectal adenoma [63]	Not reported	Case-control and prospective	Highest vs. lowest category	21	157,725	10829	RR	0.71 (0.68–0.75)	8.37e ⁻⁰⁶	62.7	No	Neither	No	0.46 1.18	III
Mortality for coronary heart disease [33]	General	Prospective	Highest vs. lowest category	18	428	1902	RR	0.818 (0.746–0.896)	0.0000171	0	No	Neither	No	0.740 0.903	III
Metabolic syndrome [61]	General	Prospective	Highest vs. lowest category	11	23,341	9140	OR	0.808 (0.719–0.908)	0.000353	66.319	Yes	Excess significance bias	Yes	0.581 1.124	III
Stroke [55]	General	Case-control	Highest vs. lowest category	16	855,672	19168	HR	0.815 (0.736–0.903)	0.0000848	64.005	No	Excess significance bias	Yes	0.578 1.148	III
Colorectal cancer [58]	Asian	Case-control	Highest vs. lowest category	14	5364	3722	OR	(0.387–0.73)	0.0000937	80.682	No	Neither	No	0.172 1.642	III
Breast cancer [50]	Women	Case-control and prospective	Highest vs. lowest category	16	782	54271	RR	0.914	0.0000584	22.012	No	Neither	No	0.841 0.995	III
All cause mortality [64]	Type 2 diabetes	Prospective	Highest vs. lowest category	3	64	1740	RR	0.859 (0.808–0.913)	0.00000121	0	No	Neither	No	0.577 1.278	III
All cause mortality [54]	Women with breast cancer	Prospective	Highest vs. lowest category	5	5512	1131	RR	0.634 (0.52–0.774)	0.00000734	0	No	Neither	No	0.459 0.876	III
Distal colorectal cancer [49]	General	Prospective	Highest vs. lowest category	8	1,541,701	2586	RR	0.704 (0.594–0.834)	0.0000499	0	No	Neither	No	0.57 0.87	III
Colorectal cancer [51]	General	Prospective	Highest vs. lowest category	22	865,829	5385	OR	0.772 (0.662–0.9)	0.000926	57.366	No	Neither	No	0.437 1.364	III
Type 2 diabetes [37]	General	Prospective	Highest vs. lowest category	9	9546	4955	RR	0.80 (0.723–0.886)	0.0000174	39.026	No	Excess significance bias	Yes	0.708 0.905	III
Renal cell carcinoma [44]	Not reported	Case-control and prospective	Highest vs. lowest category	6	935,962	5792	RR	0.841 (0.739–0.957)	0.009	23.8	No	Neither	No	0.63 1.12	III
Barrett's esophagus [46]	Mixed	Case-control	Highest vs. lowest category	3	1381	557	OR	0.423 (0.292–0.614)	5.82e ⁻⁰⁶	0	No	Neither	No	0.04 4.71	IV

(continued on next page)

Table 1 (continued)

Outcome	Population	Study design included in MA	Level of comparison	N of studies	N of participants	N of cases	Type of metric	Effect size	p-value	I ²	Publication bias	Small study effect or excess significance bias	Excess significance test	95 % CI prediction intervals	Evidence class ^a
IBD [41]	Women	Case-control	Highest vs. lowest category	2	8797	118	RR	0.579 (0.34–0.987)	0.0448	46.685	NA	NA	NA	<3 studies	IV
Breast cancer mortality [54]	Women with breast cancer	Prospective	Highest vs. lowest category	5	10280	679	RR	0.721 (0.538–0.967)	0.0288	0	No	Neither	No	0.448 1.16	IV
Type 2 diabetes [52]	Asian/US/ European population	Prospective	Highest vs. lowest category	13	143,86	27977	HR	0.905 (0.848–0.967)	0.00303	74.233	Yes	Excess significance bias	Yes	0.734 1.116	IV
Chron disease [41]	Not reported	Case-control	Highest vs. lowest category	7	172,587	855	RR	0.445 (0.288–0.687)	0.0003	56.1	Yes	Both	No	0.13 1.52	IV
Squamous esophageal cancer [46]	Mixed	Case-control	Highest vs. lowest category	6	7832	763	OR	0.525 (0.305–0.903)	0.02	82.6	No	Neither	No	0.08 3.28	IV
Inflammatory disease mortality [42]	General	Prospective	Highest vs. lowest category	3	369,932	19215	RR	0.834 (0.691–1.006)	0.06	0	No	Neither	No	0.25 2.83	NS
Hyperplastic polyps/ serrated polyps unspecified [34]	Mixed	Case-control and prospective	Highest vs. lowest category	5	39004	2411	RR	0.982 (0.932–1.033)	0.48	7	Yes	Excess significance bias	No	0.87 1.11	NS
IBD [41]	General	Case-control	Highest vs. lowest category	8	240,896	217	RR	0.671 (0.377–1.194)	0.174	74.823	No	Neither	No	0.103 4.36	NS
All-cause mortality [62]	Colorectal cancer's survivors	Prospective	Highest vs. lowest category	3	1,112,824	2081	HR	0.88 (0.542–1.427)	0.604	73.802	No	Neither	No	0.003 254.297	NS
Mortality of colorectal cancer [62]	Colorectal cancer's survivors female	Prospective	Highest vs. lowest category	2	122,878	1182	HR	0.722 (0.442–1.181)	0.194	72.544	No	NA	NA	<3 studies	NS
Proximal colorectal cancer [57]	General	Prospective	Highest vs. lowest category	7	1,149,803	2738	RR	0.928 (0.79–1.089)	0.36	0	No	Neither	No	0.752 1.145	NS
Endometrial cancer [49]	Women	Case-control and prospective	Highest vs. lowest category	13	108,778	5642	RR	0.858 (0.729–1.01)	0.0661	52.158	No	Neither	No	0.528 1.396	NS
Stroke [52]	Asian/US/ European population	Prospective	Highest vs. lowest category	2	2,102,990	474	HR	0.846 (0.623–1.149)	0.285	0	NA	NA	NA	<3 studies	NS
Prostate cancer [39]	Not reported	Case-control and prospective	Highest vs. lowest category	16	375,722	13330	RR	0.942 (0.845–1.05)	0.29	39.5	No	Neither	No	0.71 1.25	NS

Abbreviations: CVD: cardiovascular disease; CI: confidence intervals; IBD: Inflammatory bowel disease; MA: meta-analysis; OR: odds ratio; RR: relative risk; HR: hazard ratio; US: United States; NA: not available; NS: not significant.

^a Evidence class criteria: class I (convincing): statistical significance with $P < 10^{-6}$, more than 1000 cases (or >20,000 participants for continuous outcomes), the largest component study reported statistically significant effect ($P < 0.05$); 95 % prediction interval excluded the null; no large heterogeneity ($I^2 < 50\%$), no evidence of small study effects ($P > 0.10$) and excess significance bias ($P > 0.10$); class II (highly suggestive): statistical significance with $P < 10^{-6}$, more than 1000 cases (or >20,000 participants for continuous outcomes), the largest component study reported statistically significant effect ($P < 0.05$); class III (suggestive): statistical significance with $P < 10^{-3}$, more than 1000 cases (or >20,000 participants for continuous outcomes); class IV (weak): the remaining statistically significant associations with $P < 0.05$.

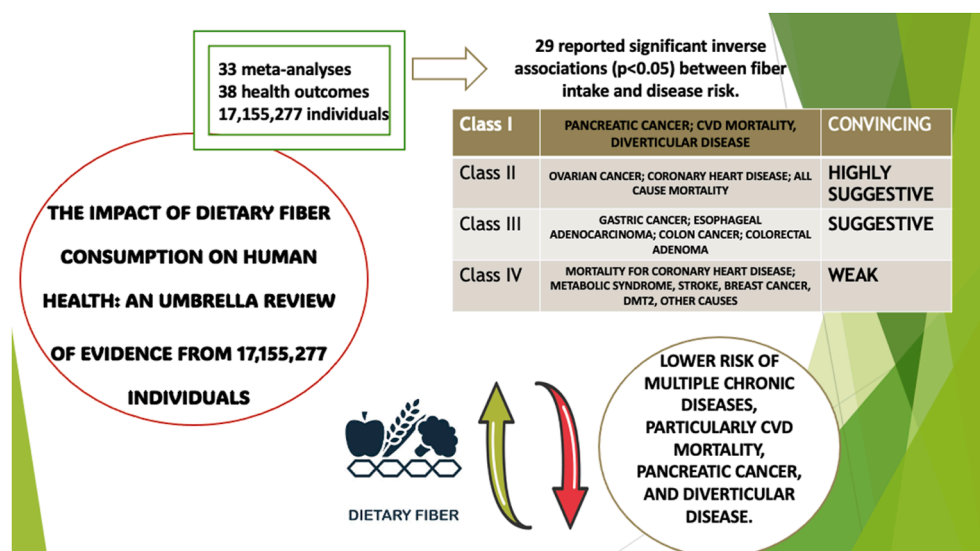


Fig. 2. Infographics.

applying rigorous credibility assessments, including small-study effects and excess significance bias. Compared to earlier findings [15], our evidence for all-cause mortality was downgraded, while diverticular disease was newly classified as having convincing evidence, reflecting the evolving strength and consistency of available data. Although prior studies have consistently reported protective associations between dietary fiber and chronic disease risk, discrepancies remain due to differences in study design (e.g., case-control vs. cohort), population characteristics, and exposure assessment methods such as FFQs. Unlike prior reviews, this work systematically evaluates 33 meta-analyses using standardized credibility grading, positioning it as a comprehensive and methodologically advanced reference. Nevertheless, further research is needed to clarify dose-response relationships, examine fiber subtypes, and reconcile variations across studies and populations [65].

Dietary fiber supports cardiovascular health through multiple mechanisms. It regulates glucose, enhance insulin sensitivity and lipid metabolism by increasing gut viscosity, slowing carbohydrate absorption which prolongs gastric emptying, and reducing post-prandial glycemic responses, all of which can also reduce risk of inflammation and malignancy [66]. Higher fiber intake lowers serum cholesterol and LDL levels by binding bile acids, promoting their excretion, and enhancing cholesterol utilization for bile acid synthesis [67]. Fiber also influences the gut microbiota, producing anti-inflammatory compounds like short-chain fatty acids (SCFAs)—key factors in reducing chronic inflammation and CVD mortality [68,69]. Emerging evidence also suggests fiber may regulate blood pressure by strengthening the endothelial function [70] and reduce thrombogenesis and platelet aggregation via cholesterol reduction [71]. For gastrointestinal health, insoluble fiber increases fecal bulk and accelerates transit time, reducing the risk of colorectal cancer and diverticulosis [66]. SCFA production from fiber fermentation strengthens the gut barrier, lowers colonic pH, and suppresses harmful microorganisms [72]. These mechanisms reduce inflammation and colonic wall weakness, lowering the risk of diverticulosis [73]. Beyond metabolic and gastrointestinal benefits, fiber is linked to a lower risk of cancers, including pancreatic and ovarian cancer. Through butyrate production, fiber acts as a histone deacetylase inhibitor, promoting histone acetylation and reactivating tumor suppressor genes such as p53 and p16—key pathways for pancreatic cancer prevention [74,75]. All of

these well-established mechanisms can explain how dietary fiber intake can enhance over-all health and thereby contributing to improved over-all mortality.

The role of dietary fiber in preventing chronic diseases has major public health implications [76,77]. Prioritizing fiber-rich, minimally processed plant foods in dietary guidelines could help reduce the global burden of non-communicable diseases. However, declining fiber intake worldwide, even in developing countries, underscores the need for urgent interventions. Policymakers should implement fiscal measures (e.g., subsidies for whole foods, taxes on ultra-processed foods) [78], reformulating processed foods to boost fiber content [79], and enhance food labeling for consumer awareness [80]. Schools, workplaces, and community settings should also be leveraged as platforms for promoting fiber-rich dietary habits from an early age [81]. Healthcare professionals play a key role in translating dietary guidelines into practice, requiring a cultural shift toward holistic lifestyle interventions—including diet, exercise, stress management, and digital tools [82]. Additionally, culinary medicine and nutrition interventions [83], along with emerging digital health tools like mobile apps and personalized nutrition platforms, present promising strategies for promoting fiber intake and tracking adherence [84]. Future research should identify the most effective public health strategies, considering socioeconomic and cultural factors that shape dietary habits.

Several limitations should be considered when interpreting the findings of this umbrella review. Firstly, as all of the included studies were observational in nature, residual confounding and reverse causality remain potential limitations. Secondly, we used established credibility assessment criteria to evaluate observational evidence [85]. While none of the individual components of these criteria can definitively prove a lack of reliability, they collectively highlight potential susceptibility to bias and uncertainty. The included meta-analyses varied in design, population, and other characteristics, so substantial heterogeneity may not always be a concern. However, we applied an $I^2 < 50\%$ threshold for class I (convincing) evidence to ensure robust associations free from bias. In cases where $I^2 > 50\%$, this usually indicates contradictory effects or non-significant associations, yet several meta-analyses classified as class II, III, or IV still had $I^2 > 50\%$, with consistent statistically significant findings across studies.

A further limitation of the umbrella review approach lies in its reliance on pre-existing meta-analyses, which have inherent limitations [86]. Meta-analytic findings can be influenced by methodological choices, such as the selection of effect estimates from primary studies, particularly when dealing with continuous and confounding risk factors (e.g., smoking, physical activity, alcohol intake). This is especially relevant when different comparisons of risk factor levels are used to express risk [87]. In our case, most of the estimates are based on the highest vs. lowest exposure analysis, for example, therefore suggesting the necessity to examine the dose-response relationships between fiber intake and health outcomes. While we initially intended to recalculate associations between dietary fiber intake and health outcomes for class I evidence by incorporating case-control studies, this was only pertinent to pancreatic cancer, as nearly all relevant studies were case-control. This is likely due to the low incidence of pancreatic cancer, which complicates the conduct of longitudinal studies. Lastly, several meta-analyses included in our review were rated as low or critically low quality according to the AMSTAR-2 assessment, indicating methodological concerns in some of the studies. These limitations must be considered when drawing conclusions from our findings.

In conclusion, this umbrella review provides strong evidence in support of dietary recommendations for higher fiber intake as a crucial element of a healthy diet. Given the widespread shortfall in fiber consumption across various populations, it is essential to implement public health strategies that encourage the consumption of whole plant foods, including vegetables, whole grains, legumes, nuts, seeds, and fruit. Clinicians should prioritize educating patients on the benefits of dietary fiber, particularly its role in reducing the risk of chronic diseases such as cardiovascular disease, cancer, and all-cause mortality, while guiding them toward sustainable dietary changes. Future research should explore the specific types of fiber and their differential effects on health outcomes, as well as investigate how fiber intake interacts with gut microbiota composition in disease prevention.

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Conflict of Interest

Marco Solmi received honoraria/has been a consultant for Angelini, AbbVie, Boehringer Ingelheim, Lundbeck, Otsuka. He is co-founder of Mesas srl, and S2M srl. The other authors did not report conflicts of interest for this work.

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NV, LF, and MS designed research; VG, CM, CDP, LC, LS and AC conducted research; NV and CG analyzed data; NV, LJD, MZ, and DN wrote the paper; LF, MB, SS, MAD critically revised the paper. NV had primary responsibility for final content. All authors read and approved the final manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnu.2025.06.021>.

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